

Assignment 2 - MDA 9159

Instructor: Dr. Guowen Huang

Fall 2025

- **This assignment is due Friday, Oct 31, 2025, at 11:55 pm.**
- You must write your answers and R code using Rmarkdown (template provided) and generate a single PDF file. **Submissions not generated by Rmarkdown will not be graded and receive zero marks.**
- Submissions must be done via Gradescope. You must carefully assign questions to their corresponding pages. Submissions without questions assigned to pages will **not** be graded. **Questions with no pages assigned to them will receive zero marks.**
- Always show all your work and add comments to your code explaining what you are doing.
- **Students are encouraged to create their own work and articulate their answers and explanations in their own words. It's essential to understand that scores are not based on the language used, as long as the core concepts and meanings are clearly conveyed. Therefore, using AI tools (like ChatGPT) to generate responses and directly copying them is not permitted. The focus of the assignment is to demonstrate your understanding, not that of the AI. This approach will also serve as valuable preparation for the upcoming closed-book midterm test.**

Dataset loaded into R by: `data(mtcars)`

Q1. Polynomial regression (curvature in horsepower) — 20 pts

Using `mtcars`, fit a quadratic model relating fuel economy to horsepower:

$$\text{mpg} \sim \text{hp} + \text{I}(\text{hp}^2).$$

- (a) Report the fitted equation with estimated coefficients.
- (b) Is the relationship **concave up** or **concave down**? Justify from the fitted quadratic term.
- (c) Compute the horsepower at which predicted MPG is maximized (the vertex).
- (d) Using a **nested-model F-test**, test whether the quadratic term improves fit over the straight line `mpg ~ hp` at $\alpha = 0.05$.

Q2. Interaction between weight and transmission — 20 pts

Let `am` be a factor (0=automatic, 1=manual). Fit the interaction model:

$$\text{mpg} \sim \text{wt} \times \text{am}.$$

- (a) Write the fitted equation.
- (b) Give the slope of `wt` for **automatic** and for **manual** cars.
- (c) Create an interaction plot with `ggplot2` showing fitted lines for each transmission group. Briefly comment on whether slopes differ meaningfully.
- (d) Using an appropriate test, assess whether the interaction is statistically significant at $\alpha = 0.05$.

Q3. Model diagnostics — 20 pts

Fit the model

$$\text{mpg} \sim \text{wt} + \text{hp} + \text{factor}(\text{cyl}).$$

- (a) Produce: Residuals-vs-Fitted, Normal Q-Q, Scale-Location, and Leverage/Cook's distance plots.
- (b) Comment on any **non-linearity**, **heteroscedasticity**, **non-normality**, or **influential** points you observe.
- (c) Without using transformations, suggest a **modeling** remedy if you detect non-linearity.

Q4. Outliers & influence — 20 pts

For the model in Q3:

- (a) Compute **leverage** (`hatvalues`), **studentized residuals** (`rstudent`), and **Cook's distance** (`cooks.distance`).

- (b) State the common **screening thresholds** you use (e.g., $h_{ii} > 2(p + 1)/n$, $|t_i| > 2$, Cook's $D > 1$).
- (c) List the top **three** potentially influential observations and show their metrics.
- (d) Refit the model after removing the **single most** influential case. Compare key coefficients and standard errors; briefly discuss any substantive changes.

Q5. Short conceptals — 20 pts

- (a) List the four standard linear-model assumptions.
- (b) Differentiate **internally studentized residuals** vs **externally studentized residuals** residuals.